Calcurion Maths Worksheets

A-Level Mathematics

Topic: Statistics – Measures of Location & Spread

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. Find the mean of $3, 7, 8, 10, 12$. **[2]**

2. Find the median of $3, 7, 8, 10, 12, 15$. **[2]**

3. Find the mode of $2, 3, 3, 5, 5, 5, 7, 8$. **[1]**

4. The mean of 5 numbers is 12. Four of the numbers are 8, 10, 14 and 15. Find the fifth. **[3]**

5. Calculate the range of $4, 9, 11, 15, 22$. **[1]**

6. Find the interquartile range of $3, 5, 7, 8, 10, 12, 15, 18, 20$. **[3]**

7. A data set has $\sum x = 120$, $n = 10$. Find the mean. **[2]**

8. A data set has $\sum x = 80$, $\sum x^2 = 800$, $n = 10$. Find the variance and standard deviation. **[4]**

9. For a frequency distribution, $\sum f = 50$, $\sum fx = 350$. Find the mean. **[2]**

10. Estimate the mean from a grouped frequency table with midpoints 5, 15, 25 and frequencies 4, 10, 6. **[4]**

11. The mean of a set of data is 20 and the standard deviation is 4. Find the mean and SD of the coded data $y = \frac{x-20}{4}$. **[3]**

12. State which measure of central tendency is most affected by extreme values. **[1]**

13. Find the 40th percentile of the ordered data: 2, 5, 7, 9, 11, 14, 18, 20, 22, 25. **[3]**

14. A data set has mean 15 and variance 9. Find $\sum x$ and $\sum x^2$ if $n = 20$. **[4]**

15. Compare the suitability of mean and median for a skewed data set. **[3]**

Section B – Hard (Q16–25)

16. Data: x: 2, 5, 7, 8, 10. Calculate the mean and the standard deviation. **[5]**

17. Two groups: Group A has mean 12, $n = 20$; Group B has mean 18, $n = 30$. Find the combined mean. **[3]**

18. A set of data has mean $\bar{x}$ and SD s. New data $y = 3x + 2$. Find the mean and SD of y in terms of $\bar{x}$ and s. **[4]**

19. Grouped data: class intervals 0–10, 10–20, 20–30 with frequencies 5, 12, 8. Estimate the mean and variance. **[6]**

20. The lower quartile is 12, median 18, upper quartile 25. Interpret the interquartile range and comment on skewness if the mean is 20. **[4]**

21. $\sum(x - \bar{x})^2 = 180$, $n = 12$. Find the standard deviation. **[3]**

22. Data coded using $y = \frac{x-50}{10}$. Mean of y is 1.2, variance of y is 0.64. Find the mean and variance of x. **[4]**

23. Find the mean and median of the stem-and-leaf data: stems 1, 2, 3 with leaves 2 5 8 — 0 1 4 7 — 3 5 (key 1—2 = 12). **[4]**

24. A data set has five numbers. The mean is 10, the median is 9, the mode is 7 and the range is 12. Construct a possible data set. **[4]**

25. Explain why the standard deviation is preferred to the range as a measure of spread. **[3]**

End of Worksheet • *Show all working clearly.*